

MTH166

Lecture-11

**Solution of Higher Order Homogeneous
LDE with Constant Coefficients-I**

Topic:

Solution of Higher order Homogeneous LDE with Constant coefficients

Learning Outcomes:

1. Solution of 3rd order (Cubic) homogeneous LDE with constant coefficients.
2. Solution of 4th order (Biquadratic) homogeneous LDE with constant coefficients.

Find the general solution of the following differential equations:

Problem 1. $y''' - 9y' = 0$

Solution: The given equation is:

$$y''' - 9y' = 0 \quad (1)$$

S.F. : $(D^3 - 9D)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(D^3 - 9D) = 0 \Rightarrow D(D^2 - 9) = 0 \Rightarrow D(D - 3)(D + 3) = 0$

$\Rightarrow D = 0, 3, -3$ (Real and distinct roots)

Let $m_1 = 0, m_2 = 3, m_3 = -3$

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} \Rightarrow y = c_1 e^{0x} + c_2 e^{3x} + c_3 e^{-3x}$$

$\Rightarrow y = c_1 + c_2 e^{3x} + c_3 e^{-3x}$ **Answer.**

Problem 2. $3y''' - 2y'' - 3y' + 2y = 0$

Solution: The given equation is:

$$3y''' - 2y'' - 3y' + 2y = 0 \quad (1)$$

S.F. : $(3D^3 - 2D^2 - 3D + 2)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(3D^3 - 2D^2 - 3D + 2) = 0 \Rightarrow D^2(3D - 2) - 1(3D - 2) = 0$

$$\Rightarrow (3D - 2)(D^2 - 1) = 0 \Rightarrow D = 1, -1, 2/3 \quad (\text{Real and distinct roots})$$

Let $m_1 = 1, m_2 = -1, m_3 = 2/3$

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} \Rightarrow y = c_1 e^{1x} + c_2 e^{-1x} + c_3 e^{2/3 x}$$

$$\Rightarrow y = c_1 e^x + c_2 e^{-x} + c_3 e^{2/3 x} \quad \textbf{Answer.}$$

Problem 3. $y''' - 2y'' + y' = 0$

Solution: The given equation is:

$$y''' - 2y'' + y' = 0 \quad (1)$$

S.F. : $(D^3 - 2D^2 + D)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(D^3 - 2D^2 + D) = 0 \Rightarrow D(D^2 - 2D + 1) = 0$

$\Rightarrow D(D - 1)^2 = 0 \Rightarrow D = 0, 1, 1$ (Two equal and one distinct real roots)

Let $m_1 = 0, m_2 = 1, m_3 = 1$

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{m_1 x} + (c_2 + c_3 x) e^{m_2 x} \Rightarrow y = c_1 e^{0x} + (c_2 + c_3 x) e^{1x}$$

$\Rightarrow y = c_1 + (c_2 + c_3 x) e^x$ **Answer.**

Polling Question

The roots of the equation: $y''' - 16y' = 0$ are:

(A) 4, 4

(B) 4, -4

(C) 0, 4, -4

(D) 0, 16

Problem 4. $27y''' - 27y'' + 9y' - y = 0$

Solution: The given equation is:

$$27y''' - 27y'' + 9y' - y = 0 \quad (1)$$

S.F. : $(27D^3 - 27D^2 + 9D - 1)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(27D^3 - 27D^2 + 9D - 1) = 0$ $[a^3 - b^3 - 3ab(a - b) = (a - b)^3]$

$$\Rightarrow (3D - 1)^3 = 0 \Rightarrow D = 1/3, 1/3, 1/3 \quad (\text{All real and equal roots})$$

Let $m_1 = 1/3, m_2 = 1/3, m_3 = 1/3$

$\therefore$ General Solution of equation (1) is given by:

$$y = (c_1 + c_2x + c_3x^2)e^{m_1x}$$

$$\Rightarrow y = (c_1 + c_2x + c_3x^2)e^{1/3x} \quad \textbf{Answer.}$$

Problem 5. $y''' - 2y'' + 4y' - 8y = 0$

Solution: The given equation is:

$$y''' - 2y'' + 4y' - 8y = 0 \quad (1)$$

S.F. : $(D^3 - 2D^2 + 4D - 8)y = 0$ where $D \equiv \frac{d}{dx}$

$$\text{A.E. :} (D^3 - 2D^2 + 4D - 8) = 0 \Rightarrow D^2(D - 2) + 4(D - 2) = 0$$

$$\Rightarrow (D - 2)(D^2 + 4) = 0 \Rightarrow D = 2, \pm 2i \quad (\text{one real and two complex roots})$$

Let $m_1 = 2, m_2 = 0 + 2i, m_3 = 0 - 2i$ [Complex roots: $(\alpha \pm i\beta)$]

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{m_1 x} + e^{\alpha x} (c_2 \cos \beta x + c_3 \sin \beta x)$$

$$\Rightarrow y = c_1 e^{2x} + e^{0x} (c_2 \cos 2x + c_3 \sin 2x)$$

$$\Rightarrow y = c_1 e^{2x} + (c_2 \cos 2x + c_3 \sin 2x) \quad \text{Answer.}$$

Problem 6. $y^{IV} - 13y'' + 36y = 0$

Solution: The given equation is:

$$y^{IV} - 13y'' + 36y = 0 \quad (1)$$

S.F. : $(D^4 - 13D^2 + 36)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(D^4 - 13D^2 + 36) = 0$

$$\Rightarrow (D^2 - 4)(D^2 - 9) = 0 \Rightarrow D = 2, -2, 3, -3 \quad (\text{real and distinct roots})$$

Let $m_1 = 2, m_2 = -2, m_3 = 3, m_4 = -3$

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + c_4 e^{m_4 x}$$

$$\Rightarrow y = c_1 e^{2x} + c_2 e^{-2x} + c_3 e^{3x} + c_4 e^{-3x} \quad \textbf{Answer.}$$

Problem 7. $y^{IV} + 8y'' - 9y = 0$

Solution: The given equation is:

$$y^{IV} + 8y'' - 9y = 0 \quad (1)$$

S.F. : $(D^4 + 8D^2 - 9)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(D^4 + 8D^2 - 9) = 0$

$$\Rightarrow (D^2 - 1)(D^2 + 9) = 0 \Rightarrow D = 1, -1, 3i, -3i \quad (\text{Mix of real and complex roots})$$

Let $m_1 = 1, m_2 = -1, m_3 = 0 + 3i, m_4 = 0 - 3i$

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{1x} + c_2 e^{-1x} + e^{0x} (c_3 \cos 3x + c_4 \sin 3x)$$

$$\Rightarrow y = c_1 e^x + c_2 e^{-x} + (c_3 \cos 3x + c_4 \sin 3x) \quad \textbf{Answer.}$$

Problem 8. $4y^{IV} + 101y'' + 25y = 0$

Solution: The given equation is:

$$4y^{IV} + 101y'' + 25y = 0 \quad (1)$$

S.F. : $(4D^4 + 101D^2 + 25)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(4D^4 + 101D^2 + 25) = 0$

$$\Rightarrow (4D^2 + 1)(D^2 + 25) = 0 \Rightarrow D = \frac{1}{2}i, -\frac{1}{2}i, 5i, -5i \quad (\text{Two sets of complex roots})$$

Let $m_1 = 0 + \frac{1}{2}i, m_2 = 0 - \frac{1}{2}i, m_3 = 0 + 5i, m_4 = 0 - 5i$ $[(\alpha \pm i\beta), (\gamma \pm i\delta)]$

$\therefore$ General Solution of equation (1) is given by:

$$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x) + e^{\gamma x}(c_1 \cos \delta x + c_2 \sin \delta x)$$

$$\Rightarrow y = e^{0x} \left(c_1 \cos \frac{1}{2}x + c_2 \sin \frac{1}{2}x \right) + e^{0x}(c_1 \cos 5x + c_2 \sin 5x)$$

$$\Rightarrow y = \left(c_1 \cos \frac{1}{2}x + c_2 \sin \frac{1}{2}x \right) + (c_1 \cos 5x + c_2 \sin 5x) \quad \textbf{Answer.}$$

Polling Question

The roots of the equation: $y^{iv} - 25y'' + 144y = 0$ are:

(A) 3,3,4,4

(B) 3,-3,4,-4

(C) 0,3,4,6

(D) 5,5,12,12

Problem 9. $y^{IV} + 50y'' + 625y = 0$

Solution: The given equation is:

$$y^{IV} + 50y'' + 625y = 0 \quad (1)$$

S.F. : $(D^4 + 50D^2 + 625)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(D^4 + 50D^2 + 625) = 0 \Rightarrow (D^2 + 25)^2 = 0$

$\Rightarrow (D^2 + 25)(D^2 + 25) = 0 \Rightarrow D = 5i, -5i, 5i, -5i$ (Repeated complex roots)

Let $m_1 = 0 + 5i, m_2 = 0 - 5i, m_3 = 0 + 5i, m_4 = 0 - 5i$ $[(\alpha \pm i\beta), (\alpha \pm i\beta)]$

$\therefore$ General Solution of equation (1) is given by:

$$y = e^{\alpha x} [(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x]$$

$$\Rightarrow y = e^{0x} [(c_1 + c_2 x) \cos 5x + (c_3 + c_4 x) \sin 5x]$$

$$\Rightarrow y = [(c_1 + c_2 x) \cos 5x + (c_3 + c_4 x) \sin 5x] \textbf{Answer.}$$

